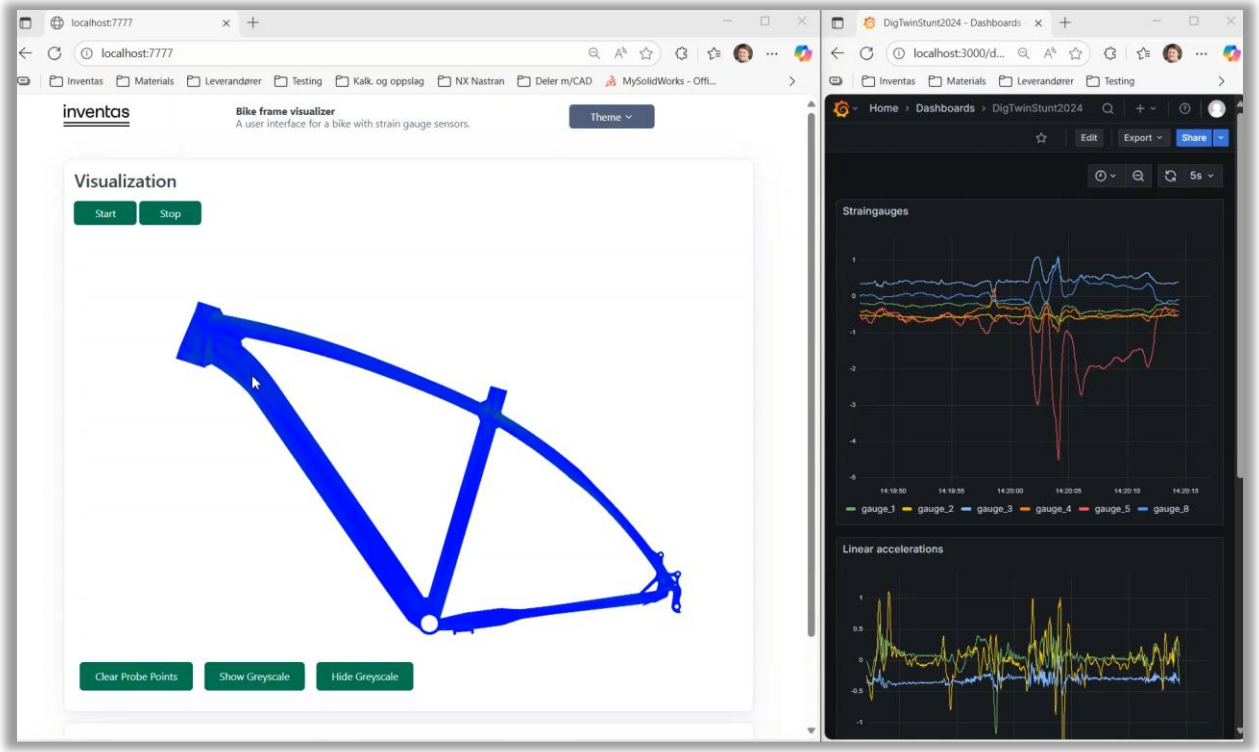
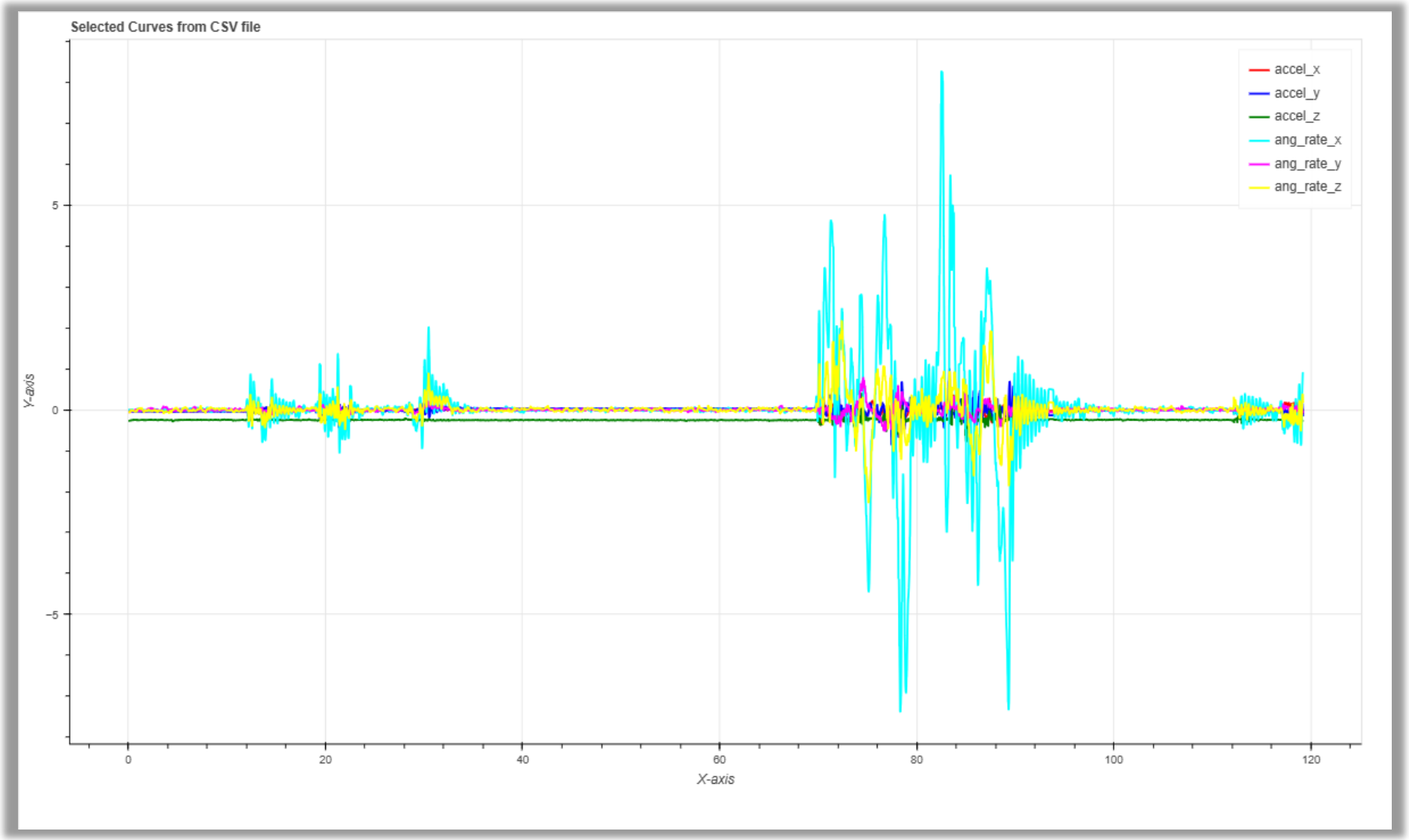
Hardrocx Frame Neural Network Pilot Study

Inputs: 6 MRU accelerations and angular rates

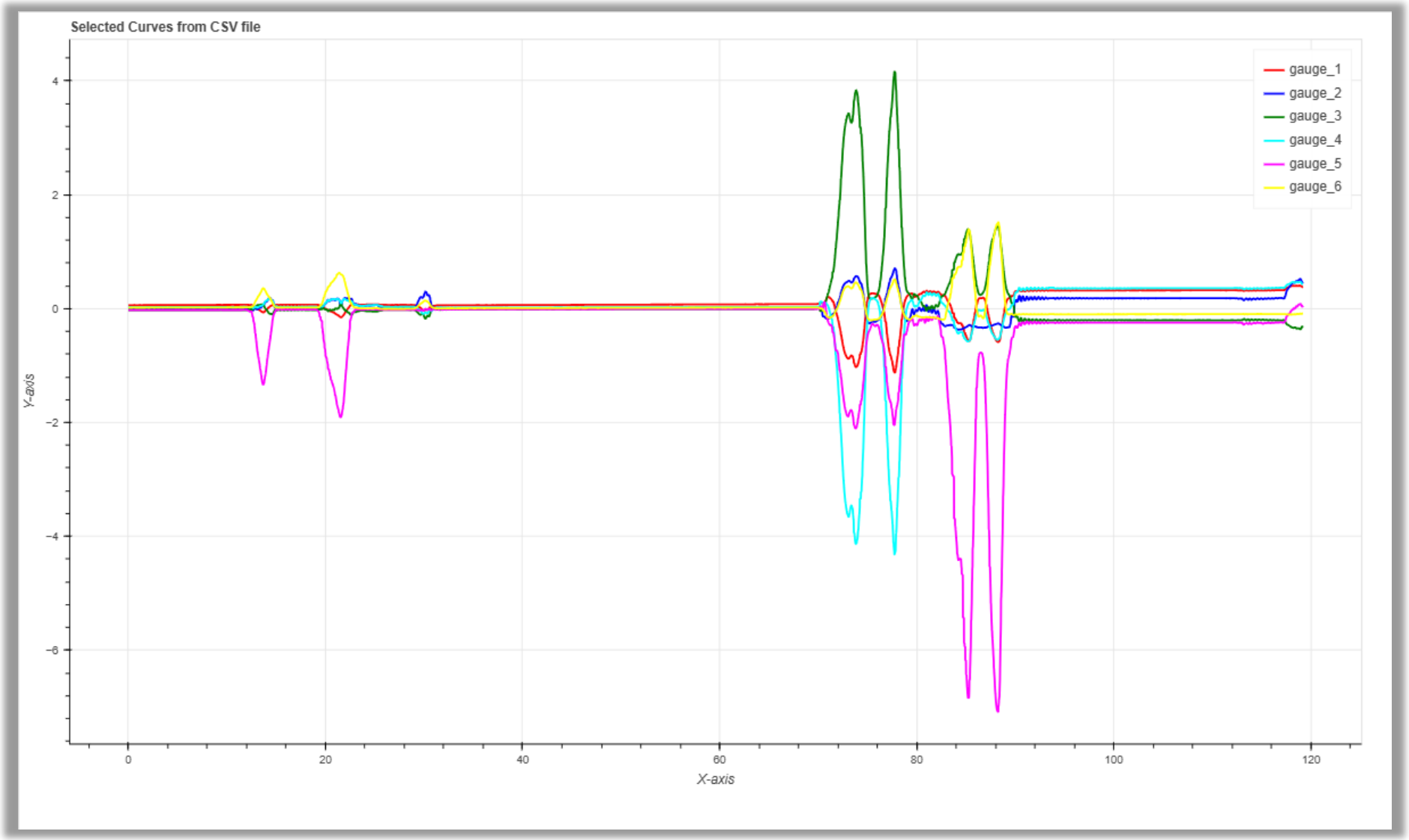
Outputs: 6 Strain Gauges located on bike frame



Measured MRU Inputs

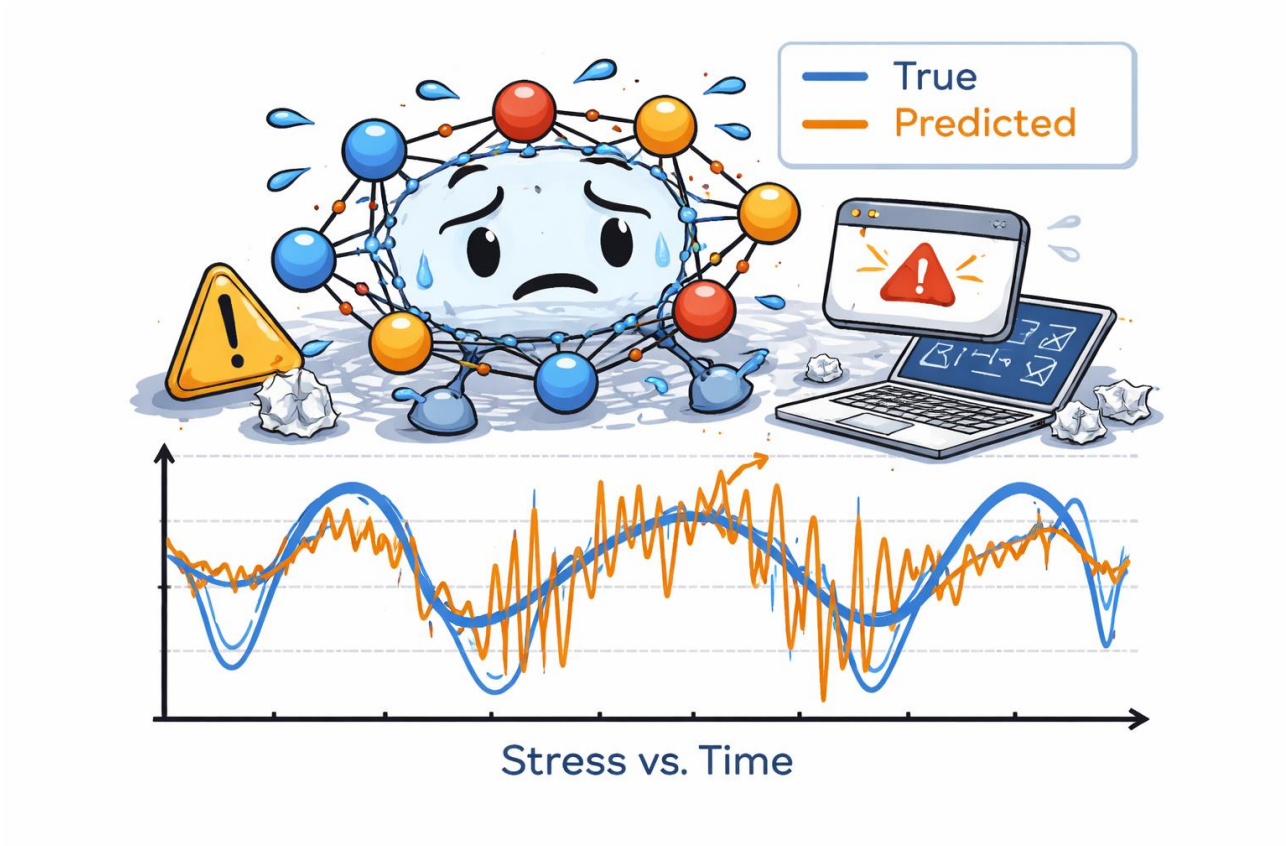


Measured Gauge Outputs

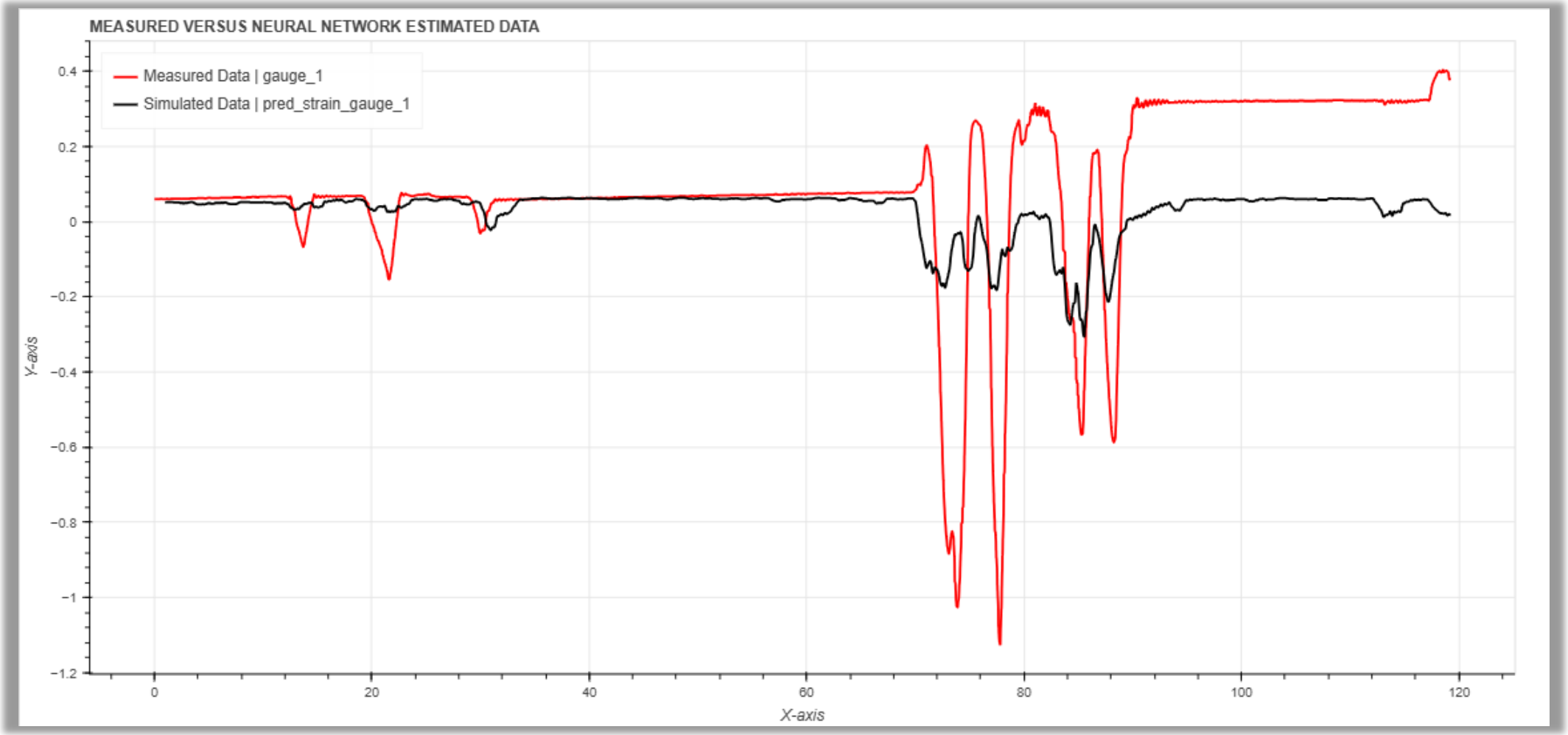


Neural Network

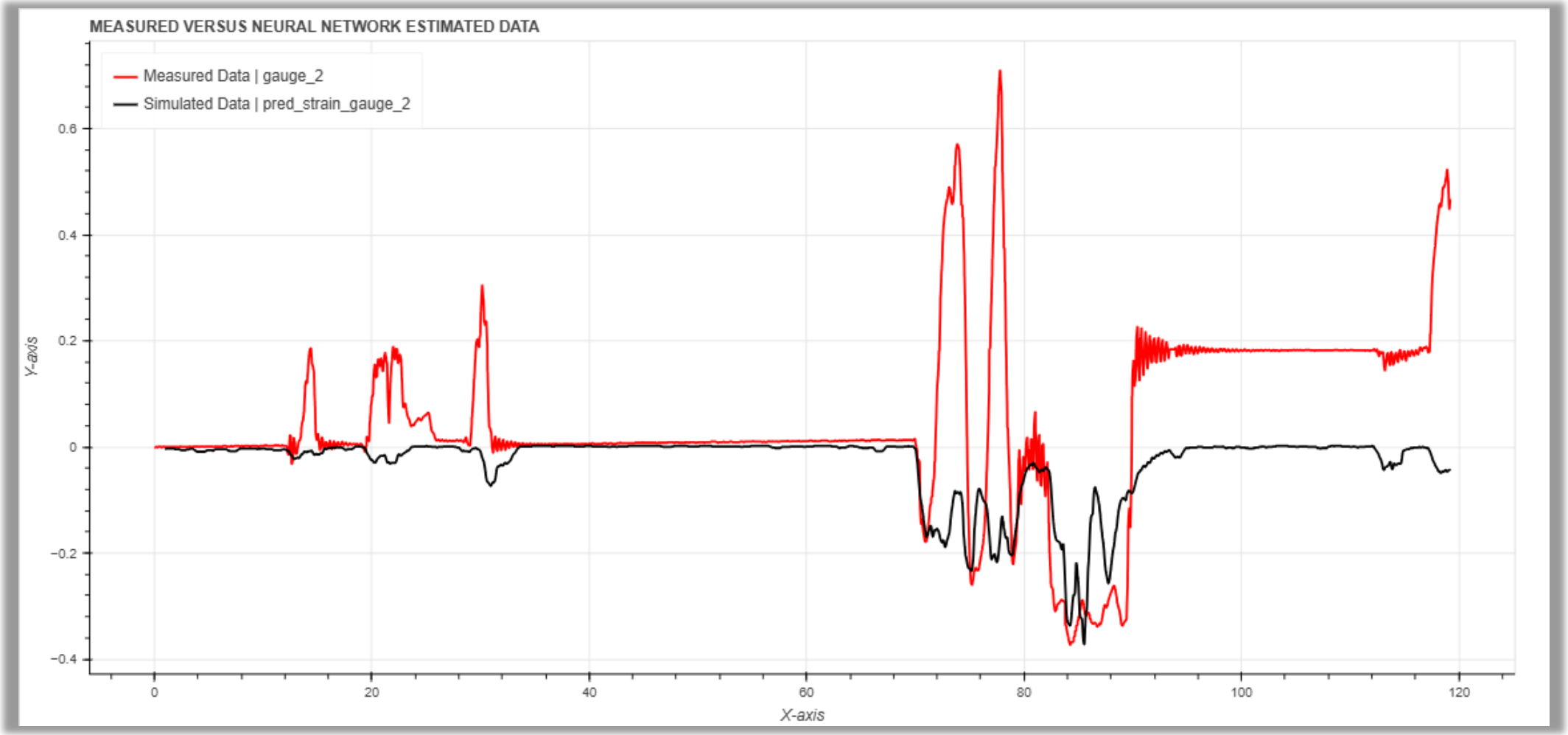
trained with default parameters



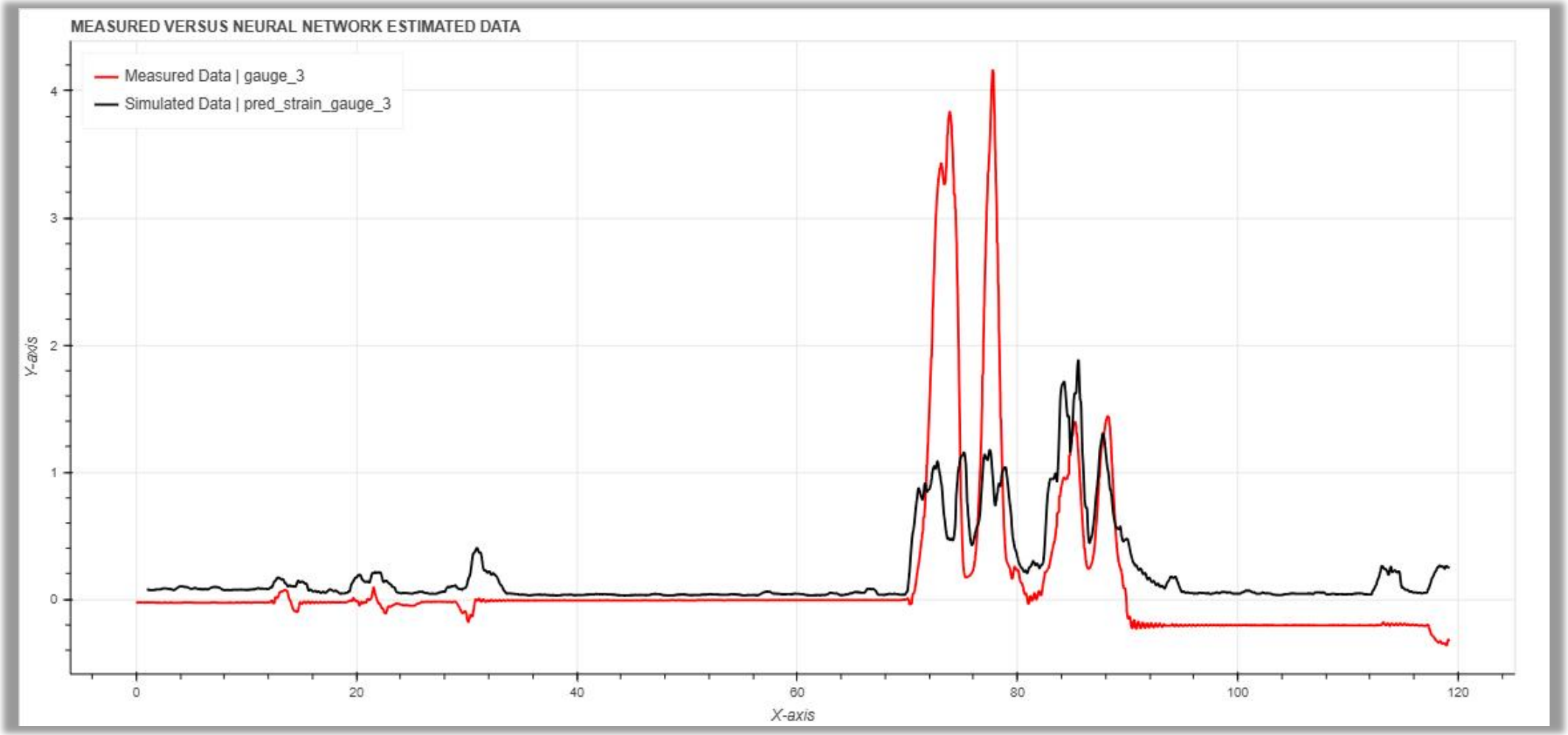
Measured versus simulated strains – Gauge 1



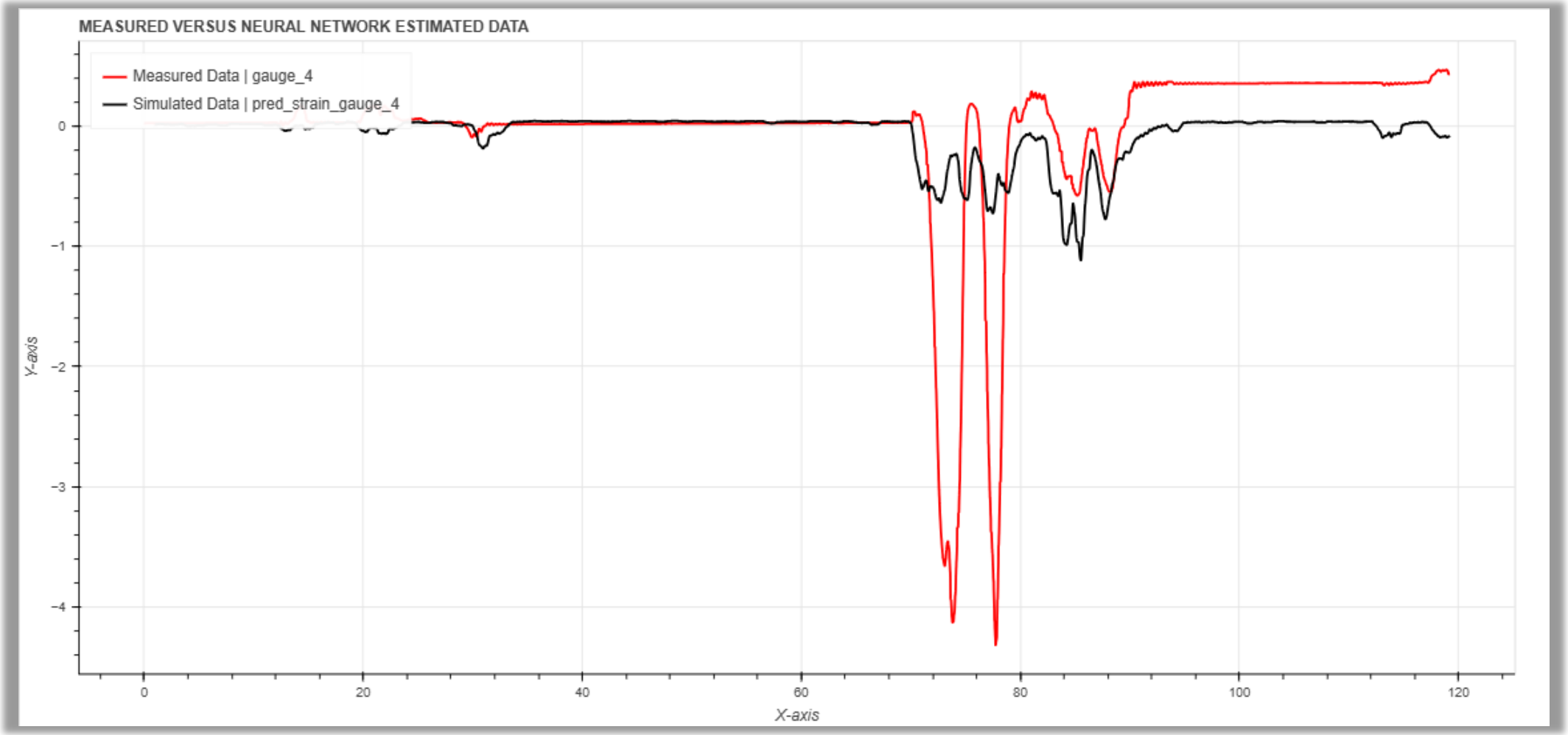
Measured versus simulated strains – Gauge 2



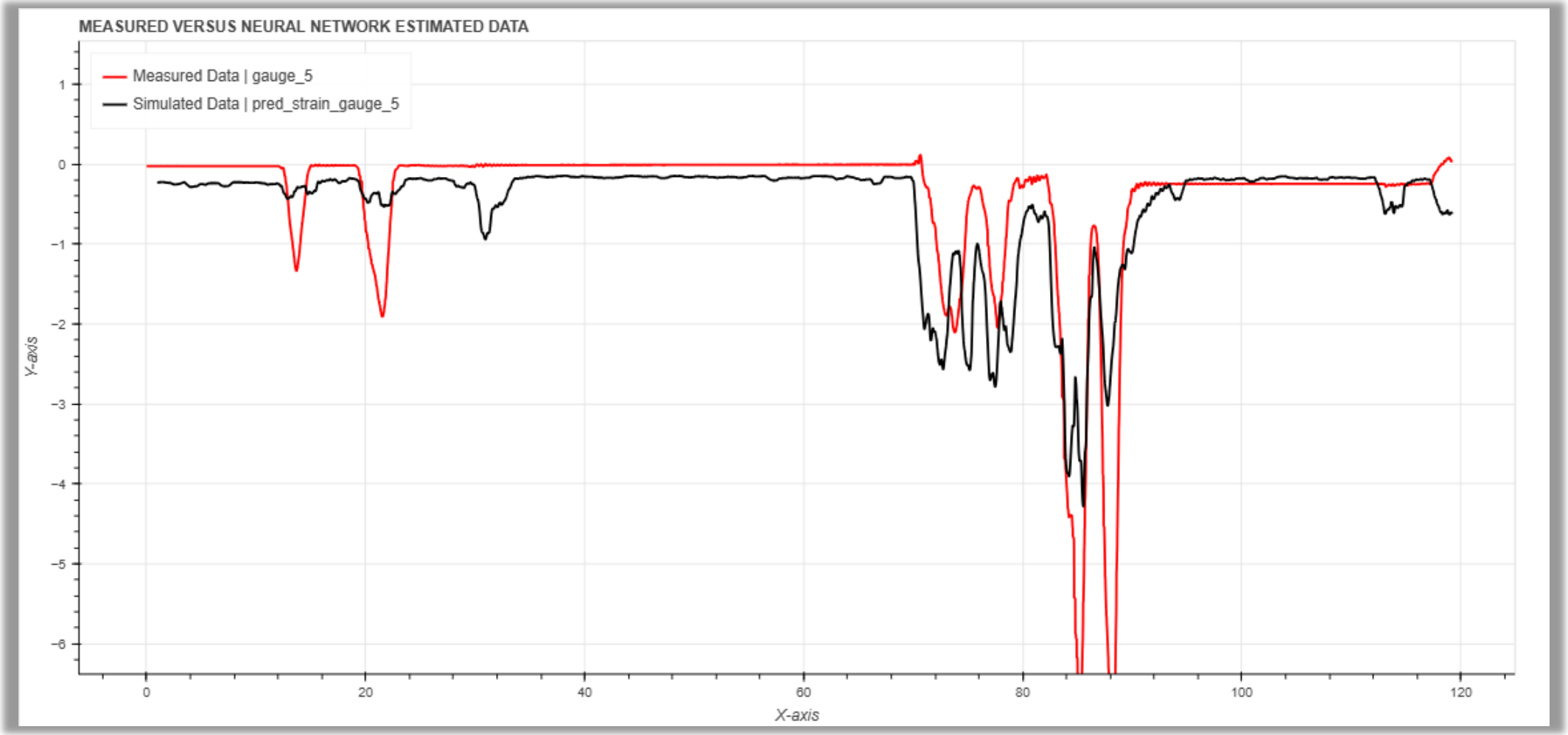
Measured versus simulated strains – Gauge 3



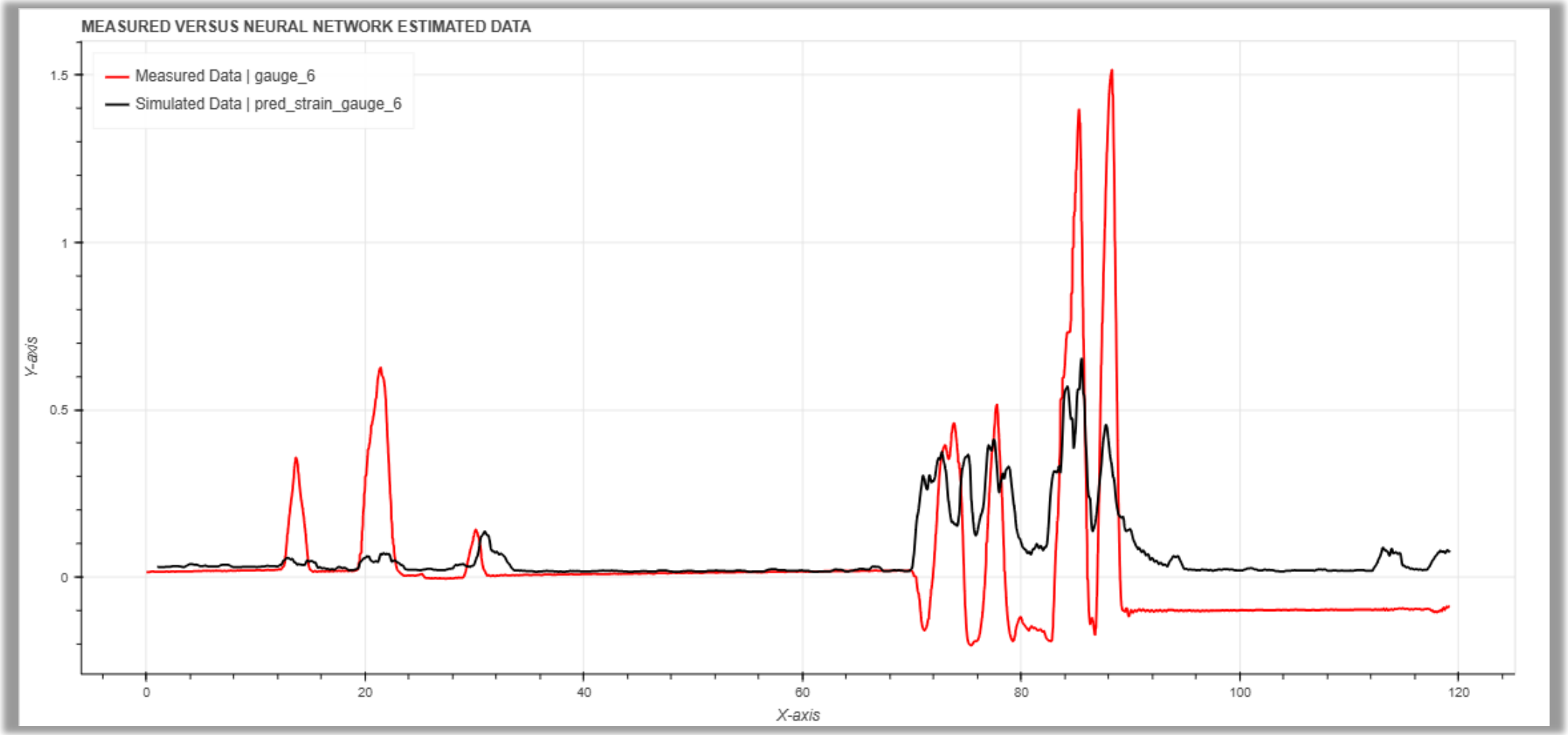
Measured versus simulated strains – Gauge 4



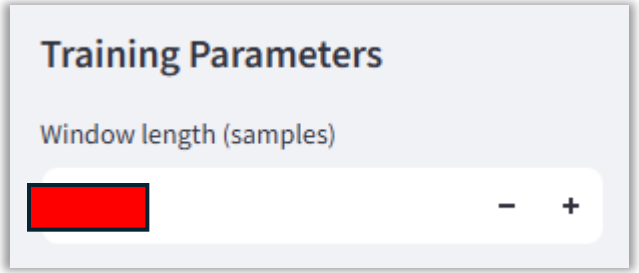
Measured versus simulated strains – Gauge 5



Measured versus simulated strains – Gauge 6



Neural Network Training – “Window Length”

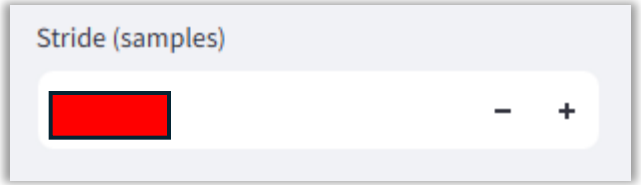


- biggest effect on **capturing physics**
- directly affects **peak accuracy AND baseline**

Vindow length (samples)		
	Too small	Too large
Effect on Accuracy	Misses event buildup	Harder to train
	Poor peak prediction	Over-smoothing
	Bad temporal context	Slower training

(samples → gives very good results on the bike data)

Neural Network Training – “Stride / steps between windows”

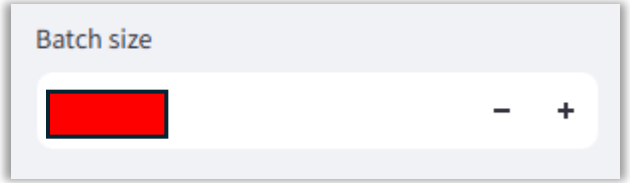


- How much you shift the window between samples.
- helps reduce those **200–300% error windows**
- big effect on **generalization**
- stride = 1 → every sample
- stride = 10 → every 0.1 s (100 Hz sampling)

Vindow length (samples)		
	Small stride	Large stride
Effect on Accuracy	more training data	less data
	better generalization	risk of missing patterns
	slower training	faster training

(stride = → gives very good results on the bike data)

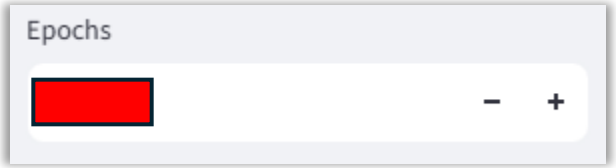
Neural Network Training – “Batch size”



- Number of samples processed before updating weigh
- Recommended 16-64 (but larger size gives less noise)

Vindow length (samples)		
	Small batch	Large batch
Effect on Accuracy	noisier updates	smoother updates
	better generalization	can overfit less
	Slower training	Faster training

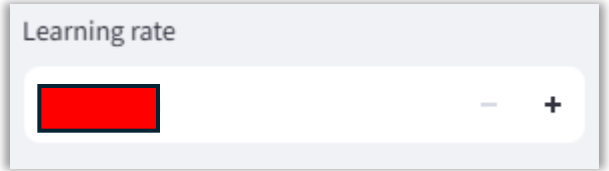
Neural Network Training – “Epochs”



- Step size when updating model weights
- big effect on overall accuracy
- especially important with small dataset
- Recommended 30-50 (but larger size gives less noise)

Vindow length (samples)		
	Too few	Too many
Effect on Accuracy	underfitting	overfitting
	poor accuracy	memorization
	fast training	slow training

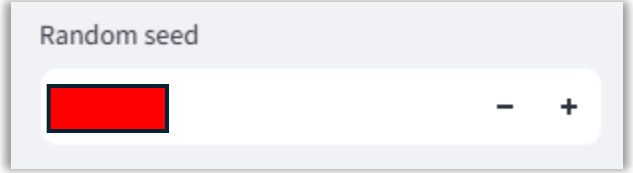
Neural Network Training – “Learning rate”



- How many times the model sees the full dataset
- big effect on training stability
- indirectly affects accuracy
- 1e-3 (0.001) → good starting point
- 1e-4 → safer, more stable

Vindow length (samples)		
	Too high	Too low
Effect on Accuracy	unstable	very slow learning
	oscillation	stuck
	bad minima	underfitting

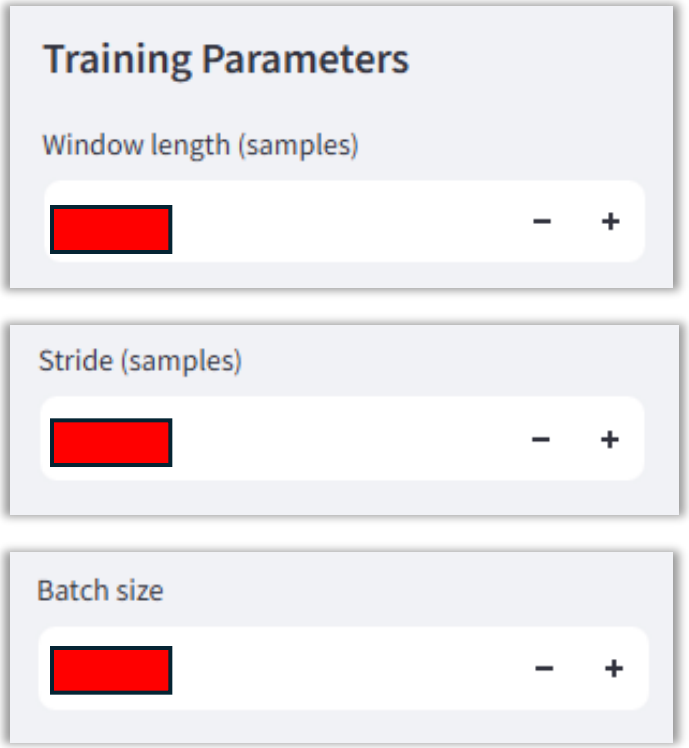
Neural Network Training – “Random seed”



- Controls weight initialization
- Can change results slightly
- Small effect on accuracy

Vindow length (samples)		
Effect on Accuracy	Too high	Too low
	Small effect on accuracy	
	can change results slightly	
	useful for debugging / comparing experiments	

Neural Network Training – Parameter Interaction



<u>These parameters interact:</u>

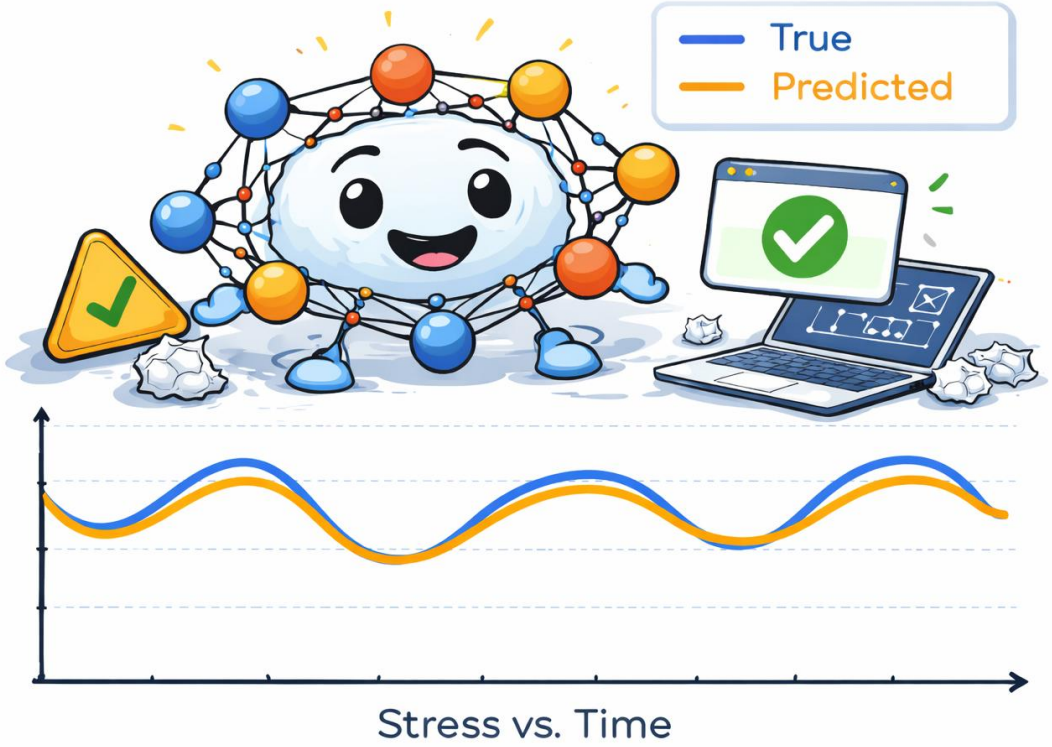
- larger window → may need smaller learning rate
- smaller stride → more data → may need fewer epochs
- small dataset → prefer:
  - small stride
  - more epochs

<u>You can try these values for bike data</u>

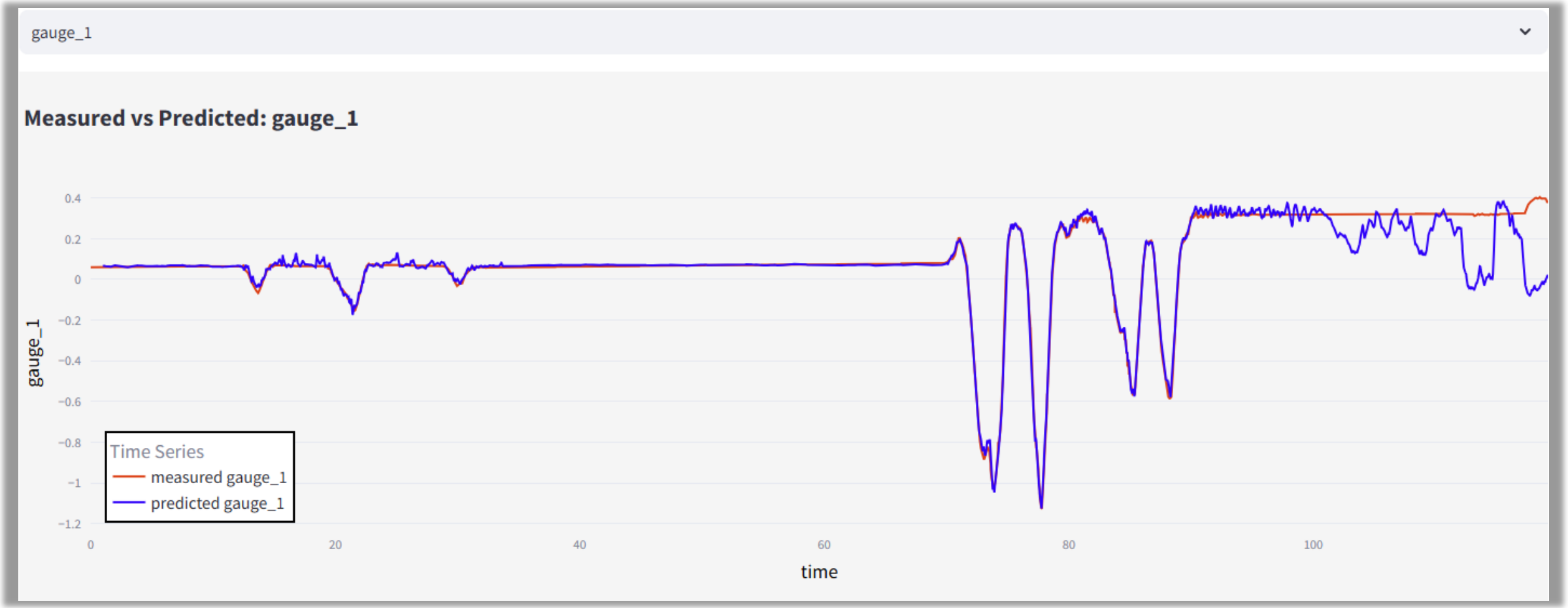
- window_length = [100, 150, 200, 250]
- strides = [1, 2, 5]
- batch_sizes = [16, 32, 64]
- epochs = [20, 50]
- learning_rates = [1e-3, 5e-4, 1e-4]

Neural Network

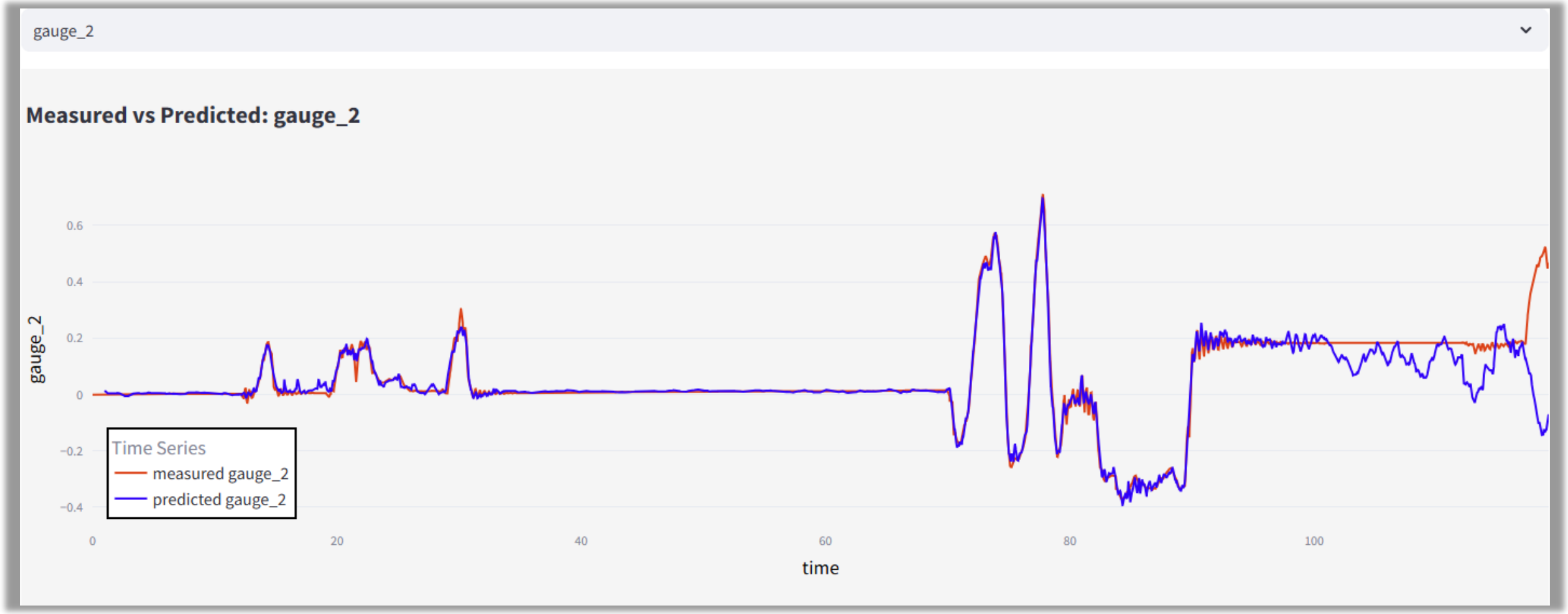
trained with good parameters



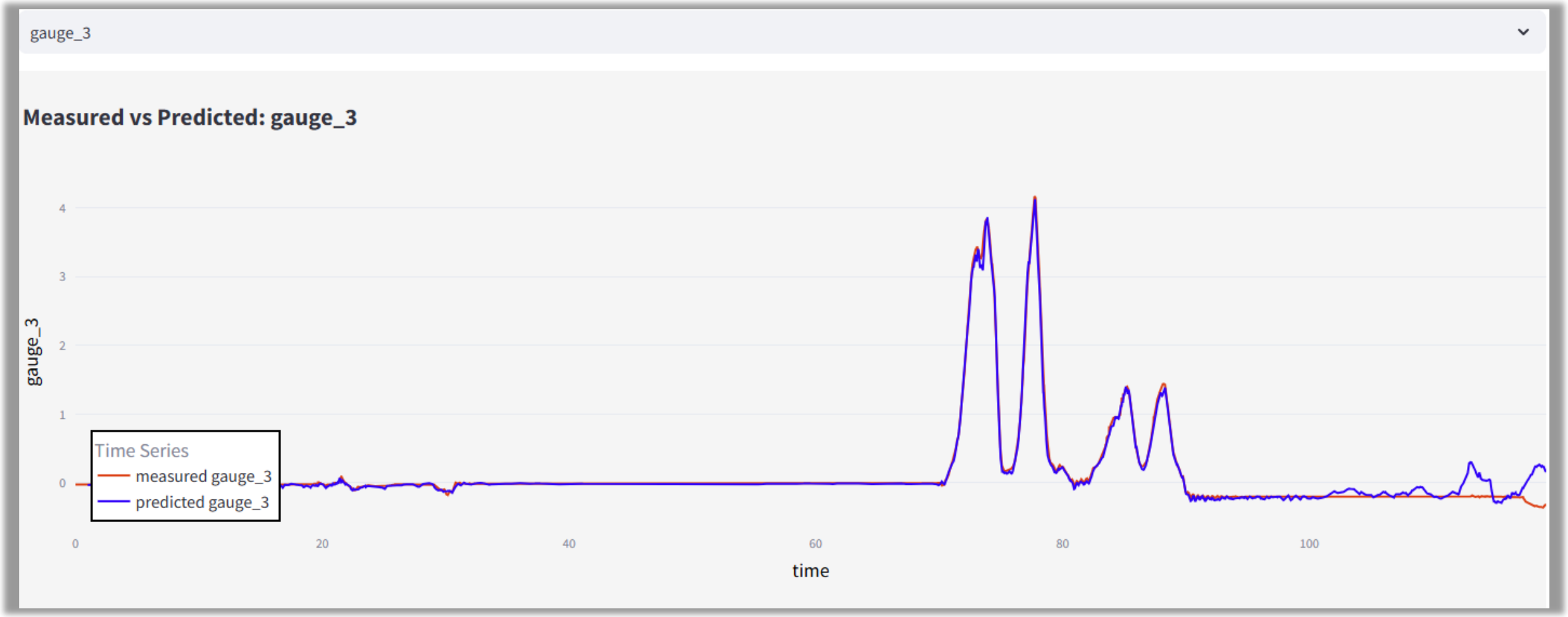
Measured versus simulated strains – Gauge 1



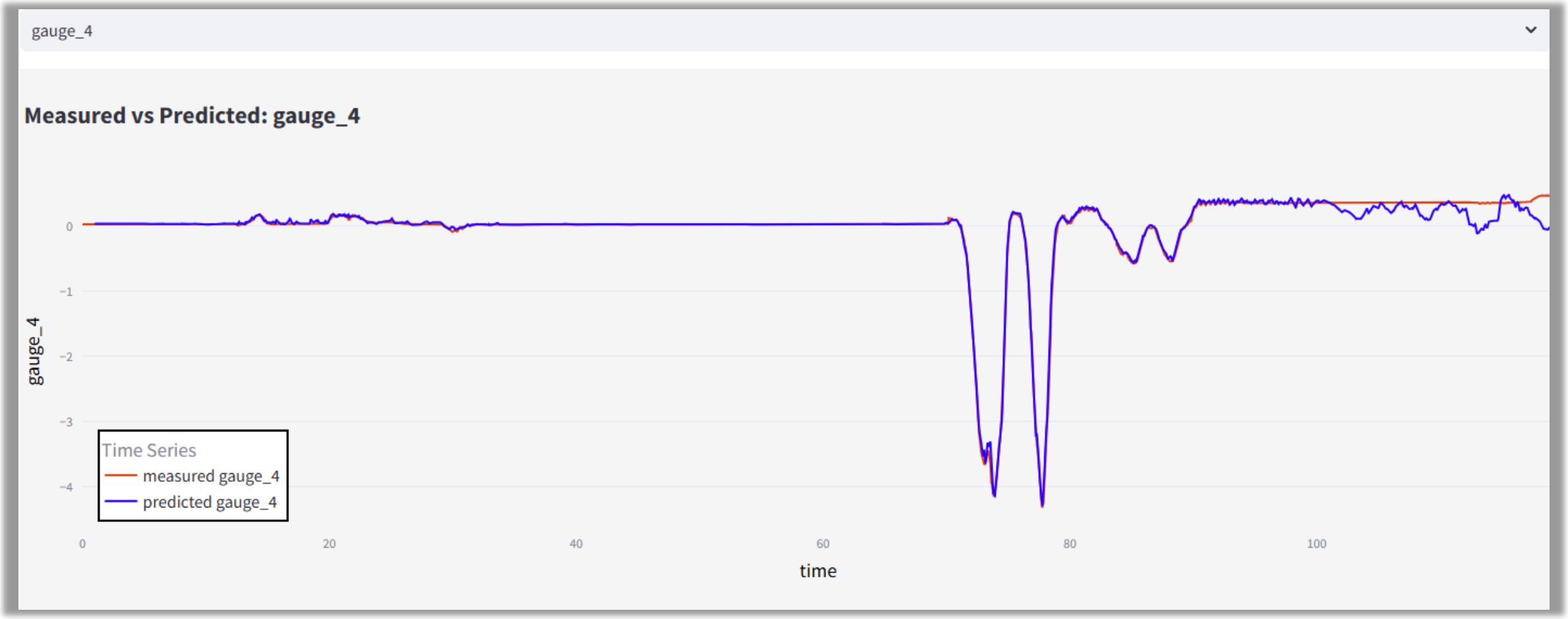
Measured versus simulated strains – Gauge 2



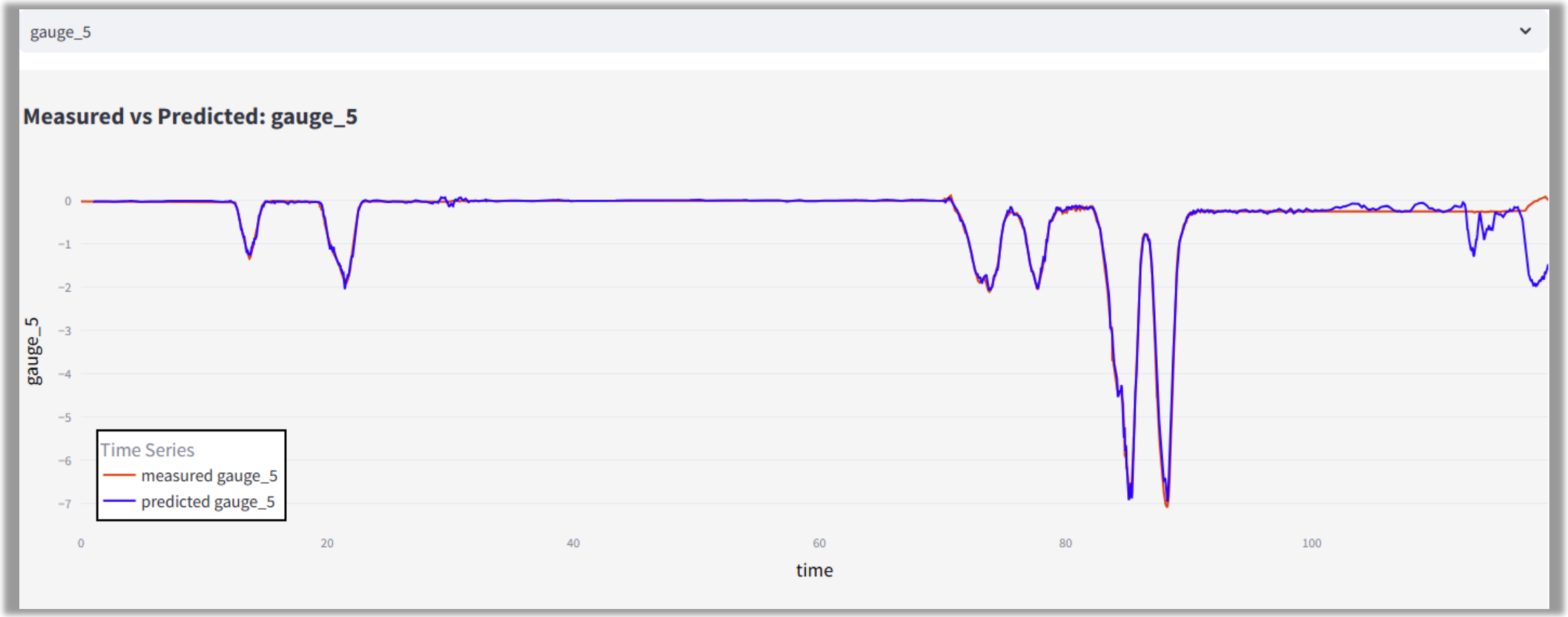
Measured versus simulated strains – Gauge 3



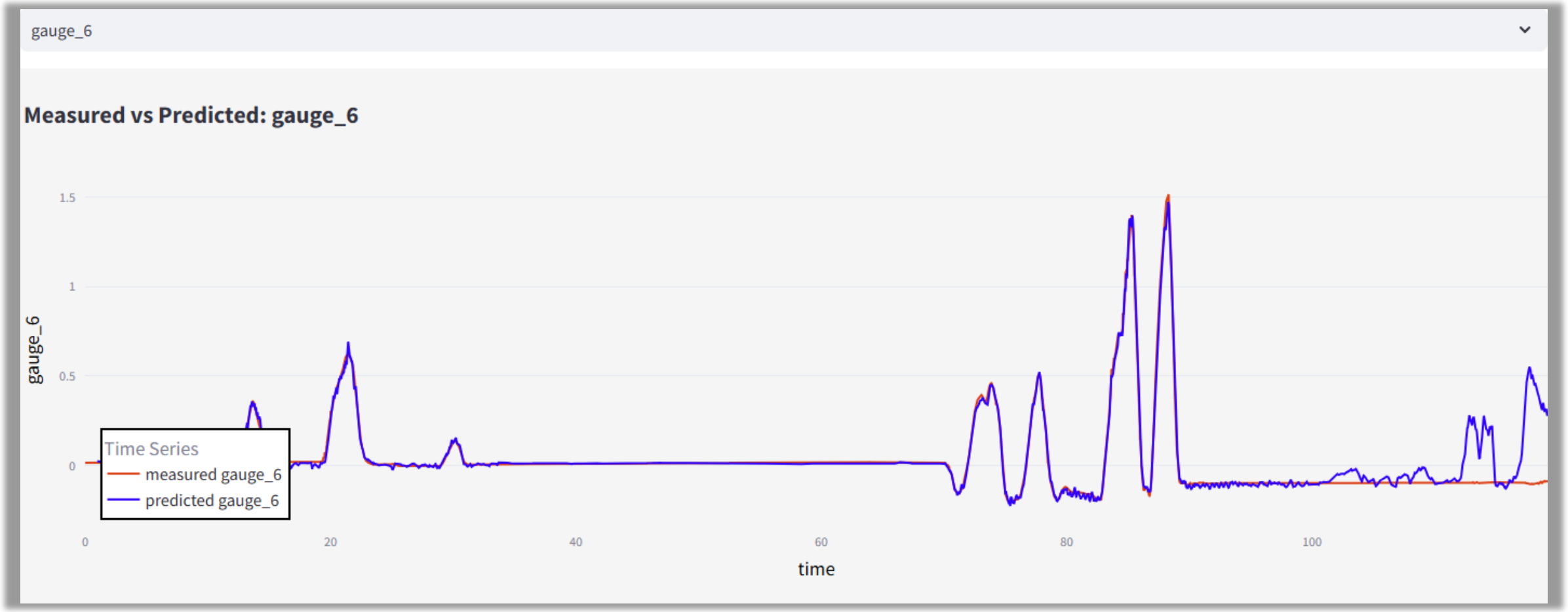
Measured versus simulated strains – Gauge 4



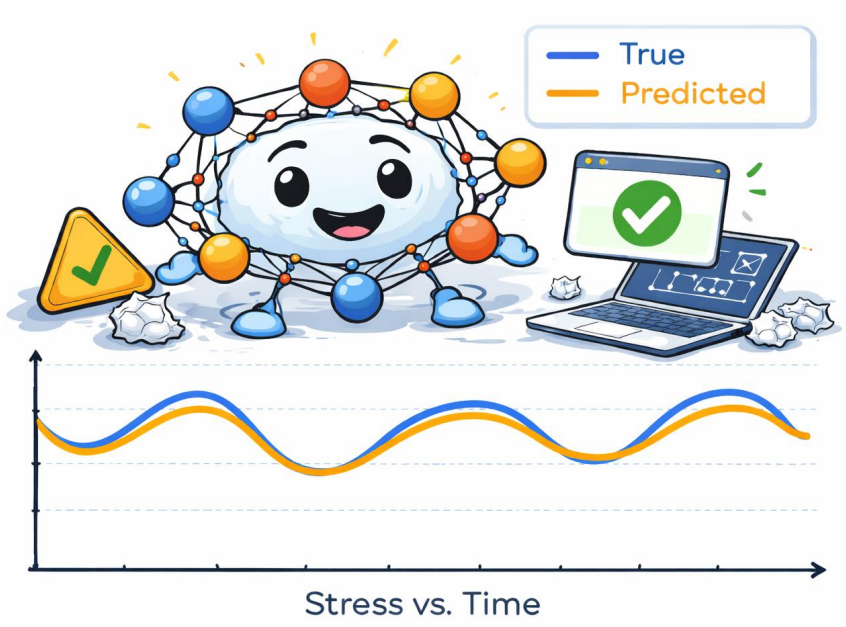
Measured versus simulated strains – Gauge 5



Measured versus simulated strains – Gauge 6



Exercise 2: Neural Network SHM network for Hardrocx bike frame:

- Train the Neural Network with optimal parameters (see recommended values on slide 17)
- Write a report based on the template on Blackboard

(**those who want can make a ROM based on FEDEM analysis instead**)